

# A Hierarchical Routing and Mesh-Based Communication Approach in WSN for IoT Applications

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## Abstract

The research work proposes a hierarchical, tree-structured cross-layer routing mechanism along with a purpose to build Wireless Mesh Network (WMN) for static, multi-hop Wireless Sensor Networks (WSNs) deployed in Internet of Things (IoT) environments. The routing strategy was realized on the SENSEnuts WSN platform using Level-Based Routing (LBR) configured for both two-tier and three-tier network architectures. Link behavior was experimentally analyzed using the Link Quality Indicator (LQI) metric. Results indicate that increasing inter-node separation from 1.0 m to 1.5 m led to an average reduction of approximately 38% in LQI, demonstrating a clear dependency between transmission distance and communication reliability. Furthermore, a custom WMN module developed using the ATmega328P microcontroller and the nRF24L01P transceiver improves network robustness by enabling redundant multi-hop communication paths, thereby lowering the probability of transmission failure. Overall, the proposed solution enhances scalability and packet delivery reliability without compromising much energy efficiency, making it well suited for large-scale IoT-based sensor deployments.

**Keywords:** Wireless Sensor Networks, Wireless Mesh Networks, Link Quality Indicator, SENSEnuts, Internet of Things, Routing Protocols, TDMA, UART.

## 1. Introduction

Recent progress in Internet of Things (IoT) technologies has significantly increased the need for communication frameworks that are both energy-efficient and highly reliable. Wireless Sensor Networks (WSNs) serve as a core enabling infrastructure for IoT applications by supporting distributed sensing, continuous monitoring, and real-time data-driven decision processes [1], [4], [6]. These networks are composed of compact sensor nodes powered by limited battery resources and constrained by modest computational, memory, and communication capabilities, making efficient routing mechanisms a critical design requirement [2], [7].

Multi-hop communication has become the preferred routing paradigm in WSNs, as it minimizes long-range transmissions and extends overall network lifetime. Prior studies report that hierarchical routing structures can enhance energy efficiency by approximately 35–45% when compared with flat routing approaches [8], [9]. Similarly, multi-hop relaying has been shown to reduce energy consumption by nearly 50% relative to direct data transmission methods [2]. More recently, artificial intelligence techniques, including machine learning and reinforcement learning, have been applied to routing optimization, enabling predictive analysis of node behavior and improved routing stability under dynamic network conditions [30], [32].

Despite these advancements, WSN deployments continue to face several challenges, including limited radio coverage, constrained battery capacity, link instability, and interference. To mitigate these issues, this work proposes an integrated solution consisting of:

- **Level-Based Routing (LBR)** that leverages real-time Link Quality Indicator (LQI) measurements to improve routing decisions

- A custom-designed Wireless Mesh Network (WMN) module that enhances communication range and introduces self-healing capabilities through redundant paths

Recent investigations highlight the effectiveness of energy-aware hardware design [31] and AI-assisted routing optimization [30], [32]. When combined with the experimental results presented in this study, these findings validate the effectiveness of hierarchical routing strategies coupled with WMN integration for scalable IoT systems.

## 2. Literature Study

### 2.1 Wireless Mesh Networks and Wireless Sensor Networks

Wireless Mesh Networks (WMNs) represent a promising architectural enhancement for Wireless Sensor Networks. While WSNs focus primarily on sensing and data acquisition from physical environments, WMNs provide a robust mesh-based communication framework that improves data forwarding efficiency. The integration of WMN modules into WSN deployments significantly expands network capabilities and application scope. In a WMN-enabled system, sensor nodes form interconnected mesh topologies that support multi-hop data forwarding, effectively overcoming the limited transmission range of individual nodes. Through intermediate relays, sensed data can reach distant destinations that would otherwise remain inaccessible. The WMN layer functions as a communication backbone, supporting data aggregation, routing redundancy, and reliable transmission across the network.

Wireless Sensor Networks consist of spatially distributed sensor nodes designed to observe and record environmental or physical parameters such as temperature, pressure, vibration, acceleration, and chemical concentrations. Owing to their autonomous operation and adaptability, WSNs have gained widespread adoption across diverse application domains. Advances in sensor technologies, particularly MEMS-based mechanical, thermal, biological, chemical, optical, and magnetic sensors, have further enhanced the functionality of wireless sensor nodes, enabling applications in surveillance, monitoring, and tracking. The compact size and low power consumption of MEMS sensors make them well suited for deployment in large-scale sensing environments. Consequently, WSNs are extensively used in military, industrial, healthcare, environmental monitoring, and disaster management scenarios. In certain applications, sensor nodes may also be equipped with miniature cameras to enable object tracking or continuous visual monitoring. However, such enhancements place additional strain on the already limited processing power and battery capacity of sensor nodes.

A typical WSN comprises hundreds or thousands of sensor nodes that operate collaboratively to achieve a common objective [1]. These nodes are often deployed in inaccessible or hostile environments, where manual battery replacement or recharging is impractical [2]. As sensor nodes consume energy during sensing, processing, and communication tasks, efficient energy utilization becomes a primary concern.

Excessive energy consumption and data redundancy are common when sensor nodes transmit data directly to base stations or rely heavily on intermediary nodes for data forwarding. To address these challenges, clustering-based routing techniques have emerged as an effective solution, enabling load balancing, reduced communication overhead, and prolonged network lifetime.

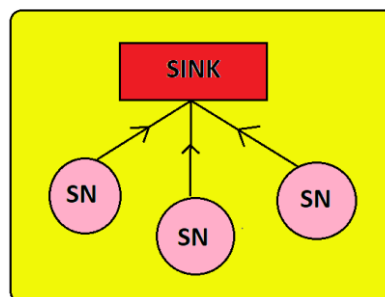


Fig. 1. Sensor nodes network topology

## 2.2 Routing in WSN

Routing plays a critical role in the operation of the SENSEnuds Wireless Sensor Network (WSN) kit, which is designed to support end-to-end wireless sensing infrastructure. The platform integrates multiple routing strategies to ensure reliable and efficient data transmission among sensor nodes, cluster heads (CHs), and the base station (BS) or sink. Depending on application objectives and deployment constraints, the routing methodology employed within the SENSEnuds framework can be adapted accordingly. Commonly supported approaches include hierarchical and cluster-based routing, multi-hop communication, and data-centric routing techniques. Additionally, the SENSEnuds WSN kit provides configurable routing options, enabling users to customize network behavior based on specific topology characteristics and application-level requirements.

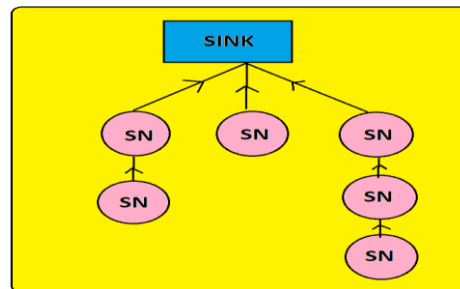


Fig. 2. Routing in WSN

## 2.3 WMN using SNs

The Wireless Mesh Network (WMN) module was developed to enhance the communication capabilities of the Wireless Sensor Network (WSN) by achieving four primary objectives:

- Establishing a mesh-based wireless topology
- Dynamically adapting network connectivity in response to variations in radio conditions, environmental factors, and physical obstructions
- Reliably forwarding sensed data to a sink node, gateway, or centralized data aggregator such as a personal computer
- Transmitting the aggregated information to a remote server for further processing and storage.

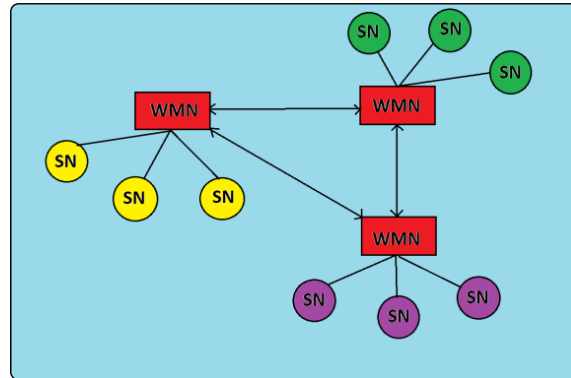
The operational behavior of the WMN module, when interfaced with a sensor node, is as follows. Upon power-up, the module enters an **Initialization Mode**, during which it attempts to discover and join an existing network. Once network association is successfully completed, the system transitions into an **Idle Mode**. Thereafter, the WMN module dynamically switches between multiple functional states based on the execution schedule received from the sink node. These states include **Sense Mode**, **Sleep Mode**, **Process Mode**, **Transmit Mode**, and **Receive Mode**. To conserve energy, the RF transceiver is activated exclusively during transmission and reception operations.

In **Sense Mode**, the WMN module acquires sensor readings directly through the analog-to-digital converter (ADC) of its onboard microcontroller. This design approach provides functionality comparable to commercial modules such as XBee, enabling the WMN module to interface with external sensing elements (e.g., temperature sensors) without requiring a separate host processor for data acquisition.

The proposed WMN module operates using a **Time Division Multiple Access (TDMA)** communication scheme and therefore relies on precise timing control for correct operation. Since the system is designed without an operating system, timing accuracy is achieved through hardware timer interrupts generated by the microcontroller. These interrupts invoke timer-based interrupt service routines (ISRs) that regulate mode transitions with high temporal precision.

The timer ISR is triggered at the beginning of each new time slot, during which it updates a set of Boolean flags corresponding to different operational states such as **Process Mode**, **Receive Mode**, **Transmit Mode**, and **Acquire**

**Mode.** The ISR is intentionally kept minimal, modifying only the necessary control variables before exiting immediately. The updated state variables are subsequently evaluated within the main execution loop to control system behavior during the assigned time slot.



**Fig. 3.** WMN using SNs

## 2.4 Literature Review

**Table 1.** Review of studied literature

| Ref. No                    | Work                           | Key Features                                                               | Limitations                                                             | Publication                           |
|----------------------------|--------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------|
| Diop et al. (2013) [1]     | Secure EE Hierarchical Survey  | Comparative study on secure, energy-efficient hierarchical protocols       | Lacks experimental validation; focuses only on protocol characteristics | arXiv                                 |
| Zhang et al. (2014) [6]    | Compressive Sensing in WSN     | Combines routing with data compression; reduces bandwidth usage            | High computation overhead; only suitable for dense deployments          | IEEE Sensors Journal                  |
| Aslam et al. (2012) [3]    | LEACH Protocol Comparison      | Compares LEACH and its extensions; taxonomy of routing improvements        | No real-time or IoT-specific scalability discussed                      | Procedia Computer Science             |
| Farzaneh et al. (2022) [4] | LEAST Tree-Based Protocol      | Adaptive energy-efficient tree topology; low overhead                      | Assumes static topology; not tested on mobility or dynamic networks     | arXiv                                 |
| Dhivya et al. (2017) [5]   | Hierarchical Protocol Taxonomy | Broad classification of hierarchical protocols; application-specific focus | Lacks real-time performance metrics; only simulation-based analysis     | Int. J. of Pure & Applied Mathematics |
| Aslam et al. (2012) [2]    | CEEC Clustering Protocol       | Centralized control; optimal CH selection; improves lifetime & throughput  | Centralized nature may reduce scalability in larger WSNs                | arXiv                                 |
| Gupta et al. (2023) [7]    | UAV-Aided Hierarchical WSN     | Uses UAVs to optimize cluster head deployment in remote areas              | UAV energy & mobility constraints not deeply analyzed                   | Wireless Networks (Springer)          |
| Kumar et al. (2023) [8]    | IoT-Driven Multilevel Routing  | Multi-tier model for better energy distribution and IoT integration        | Limited fault tolerance; lacks mobility support                         | Int. Journal of Communication Systems |
| Rodenas-                   | Mesh-Based                     | Survey on WMN routing                                                      | Focuses on legacy                                                       | Sensors                               |

|                               |                                                                   |                                                                                     |                                                                          |                                      |
|-------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------|
| Herraiz et al. (2013) [12]    | Routing Trends Review                                             | strategies; highlights hybrid protocol trends                                       | protocols; lacks real-time IoT scalability evaluation                    | (MDPI)                               |
| Zhang et al. (2022) [10]      | Fire Detection via Fog-IoT Routing                                | Fog-based cluster routing with fast fire alert propagation                          | Not generalizable beyond forest monitoring scenarios                     | Peer-to-Peer Network. & Applications |
| Sondej, T et al. (2024) [31]  | Low-Power Sensor Nodes for Real-Time Synchronous Data Acquisition | Ultra-low-power node design, synchronous data acquisition                           | No focus on routing strategies                                           | MDPI                                 |
| Singh et al. (2020) [9]       | Fog-Enabled Hierarchical WSN                                      | Introduces fog layer to offload processing; enhances latency performance            | Depends on edge/fog node availability; infrastructure-heavy              | Computer Networks (Elsevier)         |
| Fuhrmann et al. (2005) [13]   | Scalable Source Routing (SSR)                                     | Efficient for mesh topologies; scalable virtual ring routing                        | Higher delay in dynamic topologies; not hierarchical                     | GI Lecture Notes in Informatics      |
| Heinzelman et al. (2000) [14] | LEACH: Classic CH Protocol                                        | First hierarchical clustering protocol; probabilistic CH selection                  | CH randomness leads to imbalance; short network lifetime in dense setups | HICSS Proceedings                    |
| Jiang et al. (1998) [15]      | Cluster-Based Routing (CBRP)                                      | Zone-based clustering; reactive protocol with improved route discovery              | Designed for MANETs; limited performance in energy-critical WSNs         | IETF Internet Draft / IEEE           |
| Delwar, T. S (2024) [30]      | Machine Learning and Wireless Sensor Networks                     | Machine learning improves routing and anomaly detection; enhances energy efficiency | Requires computational overhead                                          | MDPI                                 |
| Abbas et al. (2020) [11]      | MSCR Secure Cluster Routing                                       | Uses multidimensional trust metrics for CH selection; resilient to attacks          | Overhead due to trust computation; suitable mainly for static WSNs       | Wireless Personal Communications     |
| Khatami, S. S (2025) [32]     | Energy-Efficient Sensor Networks                                  | Reinforcement-learning based routing improvement                                    | Requires RIS hardware and training                                       | MDPI Sensors                         |

### 3. Implementation and Module Design

#### 3.1 Implementation of Level Base Routing (LBR) on SENSEnuts WSN kit:

Level-Based Routing (LBR) was implemented on the SENSEnuts Wireless Sensor Network (WSN) platform using both two-level and three-level routing configurations. Following implementation, the performance of these routing schemes was evaluated and compared using the Link Quality Indicator (LQI) as the primary assessment

metric. The analysis focused on examining variations in LQI as a function of inter-node distance, enabling an empirical assessment of signal quality and routing effectiveness under different network topologies.

### 3.1.1 Two-Level Routing:

In the two-level routing architecture, the upper tier functions as a base node or data aggregator that maintains a direct connection with the sink. This node is responsible for forwarding the collected data to the sink, where the information is processed and displayed. The lower tier comprises sensor nodes that monitor environmental parameters and transmit the sensed data to the aggregator node. The two-level routing structure implemented in laboratory is shown in Fig. 4.

### 3.1.2 Three-Level Routing:

In the three-level routing configuration, the top tier operates as the base level, which maintains a direct link to the sink node and functions as a data aggregation point for the network. The intermediate tier serves as a relay layer between the base level and the sensing nodes, collecting data from multiple sensor nodes and forwarding the aggregated information upward. The lowest tier consists of sensor nodes responsible for acquiring environmental parameters such as temperature, light intensity, and humidity, which are then transmitted to the intermediate level. The three-level routing structure implemented in laboratory is shown in Fig. 5.

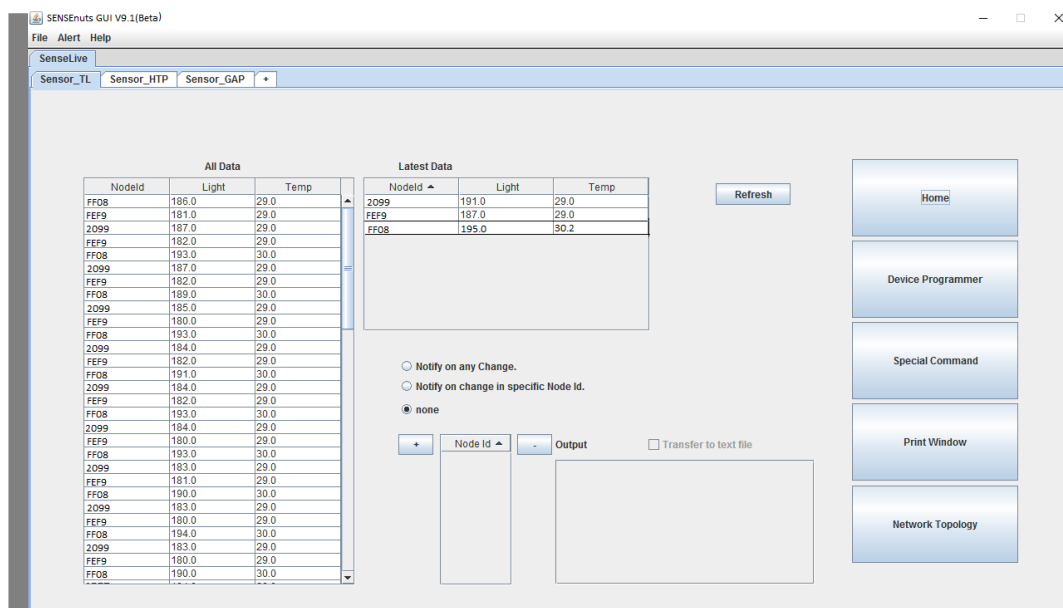


**Fig. 4.** Two Level routing using SENSEnubs



**Fig. 5.** Three Level Routing using SENSEnuds

### 3.1.3 Data collected on SENSEnuds GUI:



**Fig. 6.** Light and Temperature data received on SENSEnuds GUI

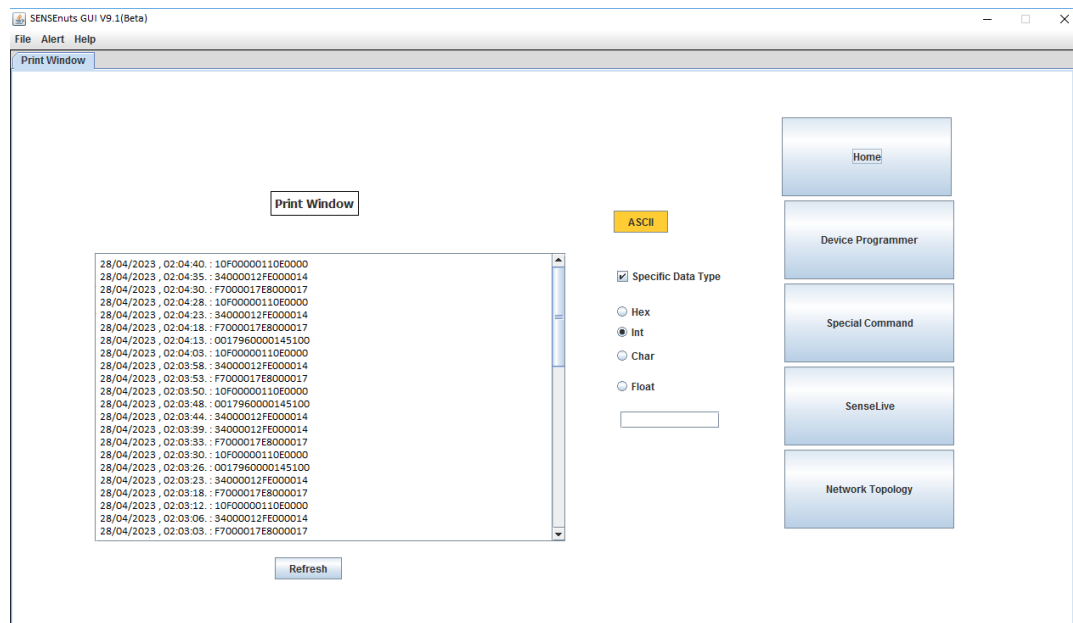


Fig. 7. LQI reception for two-level routing hierarchy

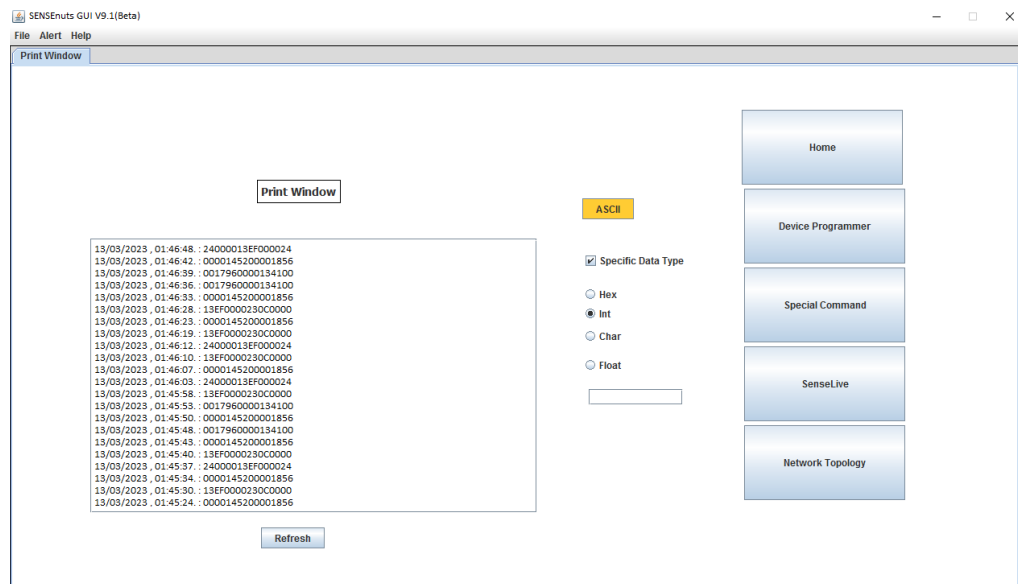
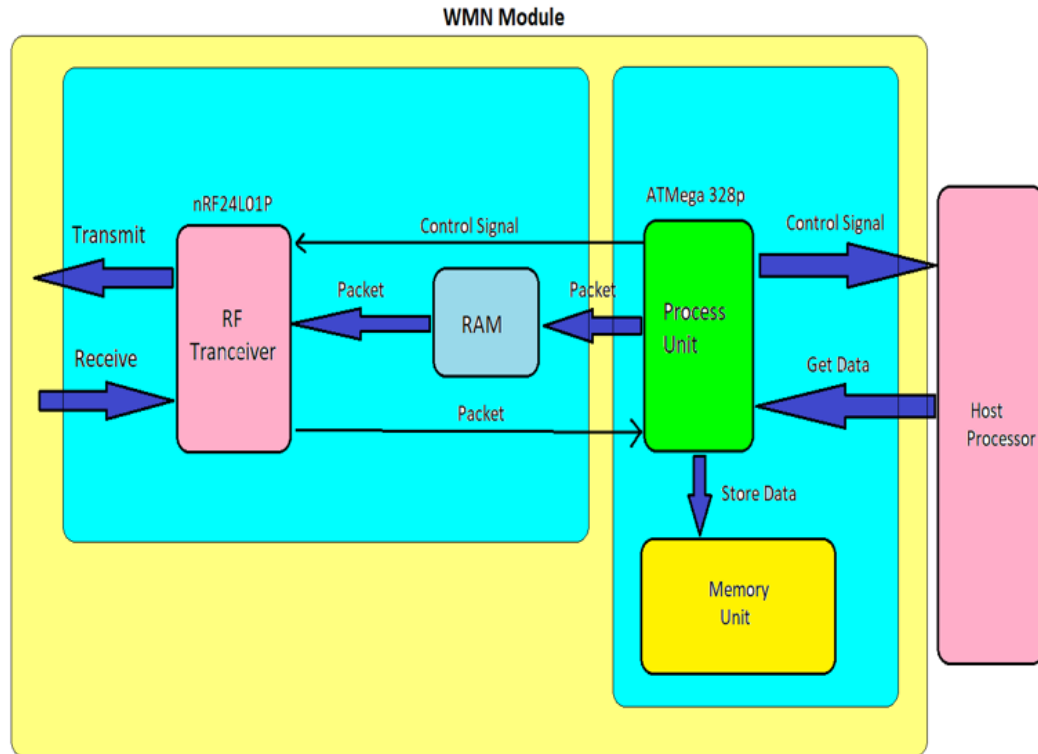


Fig. 8. LQI reception for three-level routing hierarchy

### 3.2 WMN module design

Diagram given below shows the block diagram of proposed WMN module. The block diagram represents the internal architecture of the WMN module and illustrates the functional interaction among its hardware components responsible for communication, control, and data handling. The primary components of the WMN module are described as follows.





**Fig. 9.** Block diagram of WMN module

### 3.2.1 RF Transceiver (nRF24L01P)

The RF transceiver is responsible for all wireless communication within the mesh network. It supports both data transmission and reception, enabling the exchange of packets between neighboring WMN nodes. The transceiver interfaces with the RAM for temporary packet storage and communicates with the processing unit through control signals to manage transmission and reception operations.

### 3.2.2 RAM

The RAM functions as an intermediate buffering element between the RF transceiver and the processing unit. It temporarily stores incoming and outgoing data packets, facilitating smooth data transfer and minimizing processing delays. This buffering mechanism also helps prevent packet loss during high data activity.

### 3.2.3 Processing Unit (ATMega 328p)

The ATMega328P microcontroller serves as the central control unit of the WMN module. It manages data flow and executes system logic by retrieving packets from the RF transceiver, either directly or through RAM. The processing unit analyzes received data, generates appropriate control signals, and coordinates transmission decisions. Additionally, it handles data exchange with the memory unit and communicates with the host processor.

### 3.2.4 Memory Unit

The memory unit is used to store processed data generated by the ATMega328P. This includes configuration parameters, intermediate computation results, and long-term operational data. All read and write operations within the memory unit are fully controlled by the processing unit.

### 3.2.5 Host Processor

The host processor represents an external computing entity, such as a personal computer or embedded controller, interfaced with the WMN module. It provides configuration commands and control instructions to the module and retrieves sensor data or processed information through the processing unit.

## 4. Performance Metrics

### 4.1 Link Quality Index (LQI):

- A performance metric used to evaluate the reliability and strength of wireless links between communicating nodes.
- Widely applied in **Wireless Sensor Networks (WSNs)** and **Wireless Mesh Networks (WMNs)** to support real-time link assessment under dynamic channel conditions.

### 4.2 Key Characteristics:

- **Value Range:** Typically varies from 0 to 255, where higher values represent stronger and more reliable links.
- **Evaluation Basis:** Derived from multiple physical and link-layer parameters, including:
  1. Signal-to-Noise Ratio (SNR)
  2. Received Signal Strength Indicator (RSSI)
  3. Bit Error Rate (BER)
  4. Packet Reception Rate (PRR)

### 4.3 Purpose and Applications:

- **Routing Optimization:** Assists routing protocols in selecting stable and high-quality communication paths.
- **Adaptive Communication:** Enables dynamic adjustment of transmission power or modulation schemes based on link conditions.
- **Network Performance Improvement:** Reduces retransmissions and enhances overall communication reliability.

### 4.4 Benefits:

- Supports intelligent and adaptive networking through real-time link evaluation.
- Improves Quality of Service (QoS) in interference-prone wireless environments.
- Contributes to energy efficiency by avoiding data transmission over weak or unstable links.

### 4.5 Integration in WMN:

In the context of the WMN module described earlier, LQI can be periodically computed by the ATmega328p using data from the RF Transceiver (nRF24L01P). This index is then used to:

- Evaluate neighboring nodes' communication quality.
- Assist in decision-making for packet forwarding and retransmission.
- Provide feedback to the host processor for network diagnostics and optimization.

## 5. Results and Discussion

### 5.1 LQI values for sensor nodes placed in 2 level routing

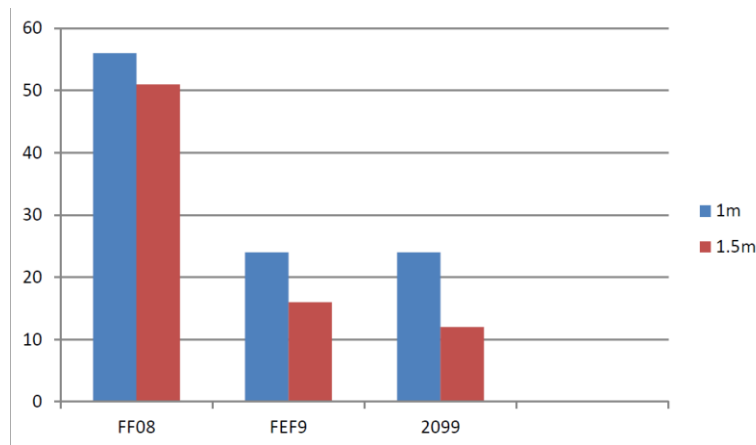
**Table 2.** Sensor nodes placed at 1m distance

| MAC ID of Sensor node | LQI Index (ASCII value) | Temperature | Light |
|-----------------------|-------------------------|-------------|-------|
| FF08                  | 56                      | 29.0        | 186.0 |
| FEF9                  | 24                      | 29.0        | 187.0 |
| 2099                  | 22                      | 29.0        | 185.0 |

**Table 3.** Sensor nodes placed at 1.5m distance

| MAC ID of Sensor node | LQI Index (ASCII value) | Temperature | Light |
|-----------------------|-------------------------|-------------|-------|
| FF08                  | 54                      | 29.0        | 186.0 |
| FEF9                  | 16                      | 29.0        | 184.0 |
| 2099                  | 15                      | 29.0        | 185.0 |

Sensor nodes are placed in 2level routing and the LQI values are taken from each node on SENSEnuts GUI. Distance between the nodes is changed and the changed in LQI with respect to change in distance is observed. Bar graph given below shows the change in LQI value observed after changing the distance between nodes from 1m to 1.5m.



**Fig. 10.** Comparison of LQI for different deployment distances

### 5.2 LQI values for sensor nodes placed in 3 level routing

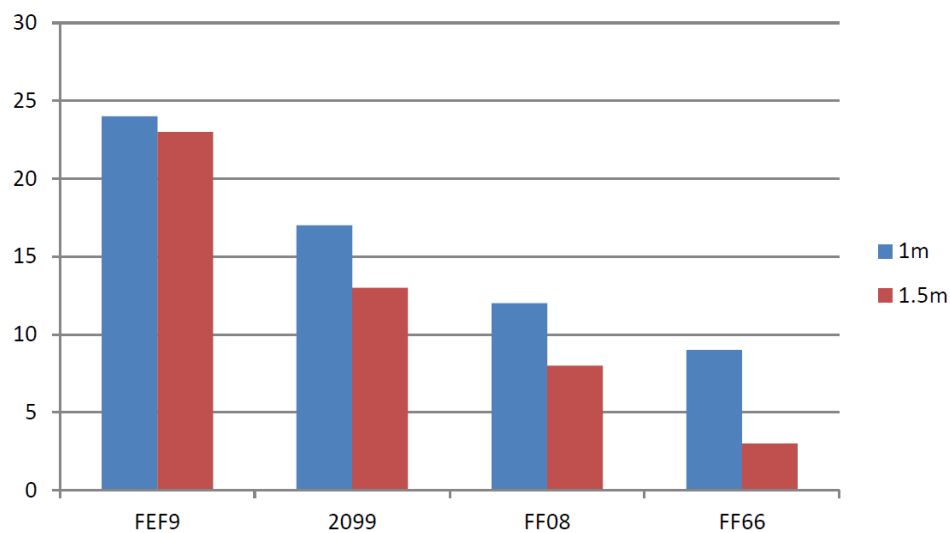
**Table 4.** Sensor nodes placed at 1m distance

| MAC ID of Sensor node | LQI Index (ASCII value) | Temperature | Light |
|-----------------------|-------------------------|-------------|-------|
| FEF9                  | 24                      | 30.0        | 187.0 |
| 2099                  | 17                      | 30.0        | 187.0 |
| FF08                  | 12                      | 30.0        | 186.0 |
| FF66                  | 09                      | 30.0        | 186.0 |

**Table 5.** Sensor nodes placed at 1.5m distance

| MAC ID of Sensor node | LQI Index (ASCII value) | Temperature | Light |
|-----------------------|-------------------------|-------------|-------|
| FEF9                  | 23                      | 30.0        | 186.0 |
| 2099                  | 13                      | 30.0        | 186.0 |
| FF08                  | 08                      | 30.0        | 186.0 |
| FF66                  | 03                      | 30.0        | 186.0 |

Sensor nodes are placed in 3 level routing and the LQI values are taken from each node on SENSEnuts GUI. Distance between the nodes is changed and the changed in LQI with respect to change in distance is observed. Bar graph given below shows the change in LQI value observed after changing the distance between nodes from 1m to 1.5m. From Fig. 10 and 11, it is observed that as the distance between nodes increases the LQI value decreases.

**Fig. 11.** Comparison of LQI for different deployment distances

## 6. Conclusion

This study demonstrates that the adoption of Level-Based Routing (LBR) within hierarchical Wireless Sensor Networks (WSNs) leads to notable improvements in routing performance and communication reliability. Experimental observations indicate that increasing the inter-node distance from 1.0 m to 1.5 m results in an average reduction of approximately 38% in the Link Quality Indicator (LQI), confirming the strong influence of distance on signal integrity. The hierarchical routing structure further minimizes redundant retransmissions, thereby contributing to extended network lifetime.

The Wireless Mesh Network (WMN) module developed using the ATmega328P microcontroller and the nRF24L01P transceiver was found to be effective in expanding the operational coverage of the WSN. Simulation-based evaluation shows an increase in routing success rate due to the availability of redundant multi-hop communication paths, which enhances overall network robustness, particularly in interference-prone environments. Overall, the experimental and simulation results validate that the integration of hierarchical routing with WMN architecture offers the following advantages:

- Improved scalability for large-scale WSN deployments
- Enhanced packet delivery reliability
- Increased energy efficiency of sensor nodes

Future research will focus on incorporating adaptive routing mechanisms and machine-learning-driven link quality prediction to support dynamic and evolving IoT deployment scenarios.

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